UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*8991003533*

**PHYSICS** **5054/04**

Paper 4 Alternative to Practical **May/June 2009**

**1 hour**

Candidates answer on the Question Paper.

No Additional Materials are required.

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **11** printed pages and **1** blank page.


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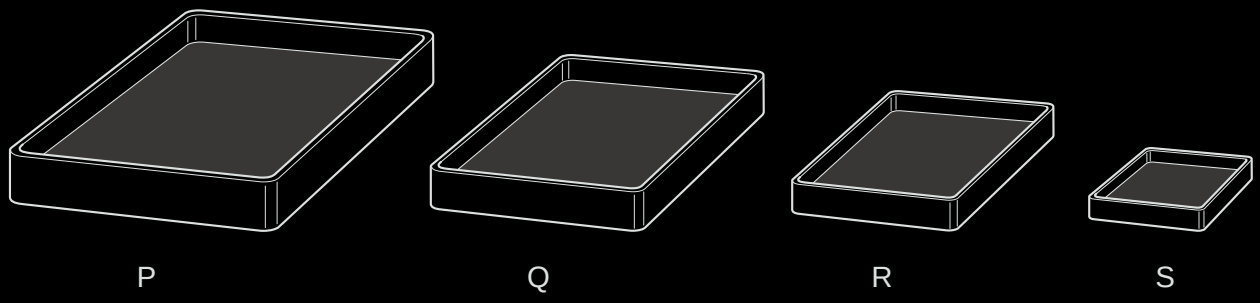



**Fig. 1.1**

A student half-fills the containers with water and leaves them for several days. He measures the total mass of each container and water on a top-pan balance at the same time each day for six days.

The following readings are obtained.

| | mass of container and water / g | | | |
|---|---|---|---|---|
| | P | Q | R | S |
| initial reading | 1378 | 573 | 196 | 129 |
| after 1 day | 1348 | 556 | 187 | 123 |
| after 2 days | 1316 | 540 | 178 | 117 |
| after 3 days | 1287 | 524 | 169 | 112 |
| after 4 days | 1253 | 509 | 159 | 105 |
| after 5 days | 1227 | 491 | 149 | 101 |
| after 6 days | 1192 | 474 | 140 | 95 |

**(a)** Complete the table below to show the loss in mass *m* after time *t* for containers P and S.

| time *t* / days | *m* / g | |
|---|---|---|
| | P | S |
| 0 | 0 | 0 |
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |

[2]




[6]

**(c)** Describe the relationship between $m$ and $t$.

.....

.....

..... [2]

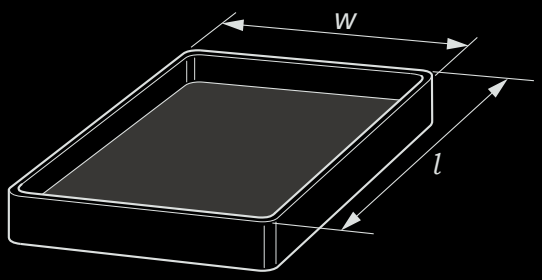



**Fig. 1.2**

An estimate of the surface area $A$ of the water is given by

$$A = l\,w.$$

**(i)** Complete the table below, giving the values of $A$ to the nearest $cm^2$.

| container | $l$ / cm | $w$ / cm | $A$ / $cm^2$ |
|---|---|---|---|
| P | 22.8 | 16.2 | |
| Q | 18.8 | 11.3 | |
| R | 13.2 | 8.9 | |
| S | 9.2 | 9.2 | |

[2]

**(ii)** By referring to Fig. 1.2, explain why $A$ is only an estimate of the surface area.

.....

..... [1]

**(iii)** The student concludes that “rate of evaporation increases with surface area”.

Explain how your graph in **(b)** supports his statement.

.....

..... [1]

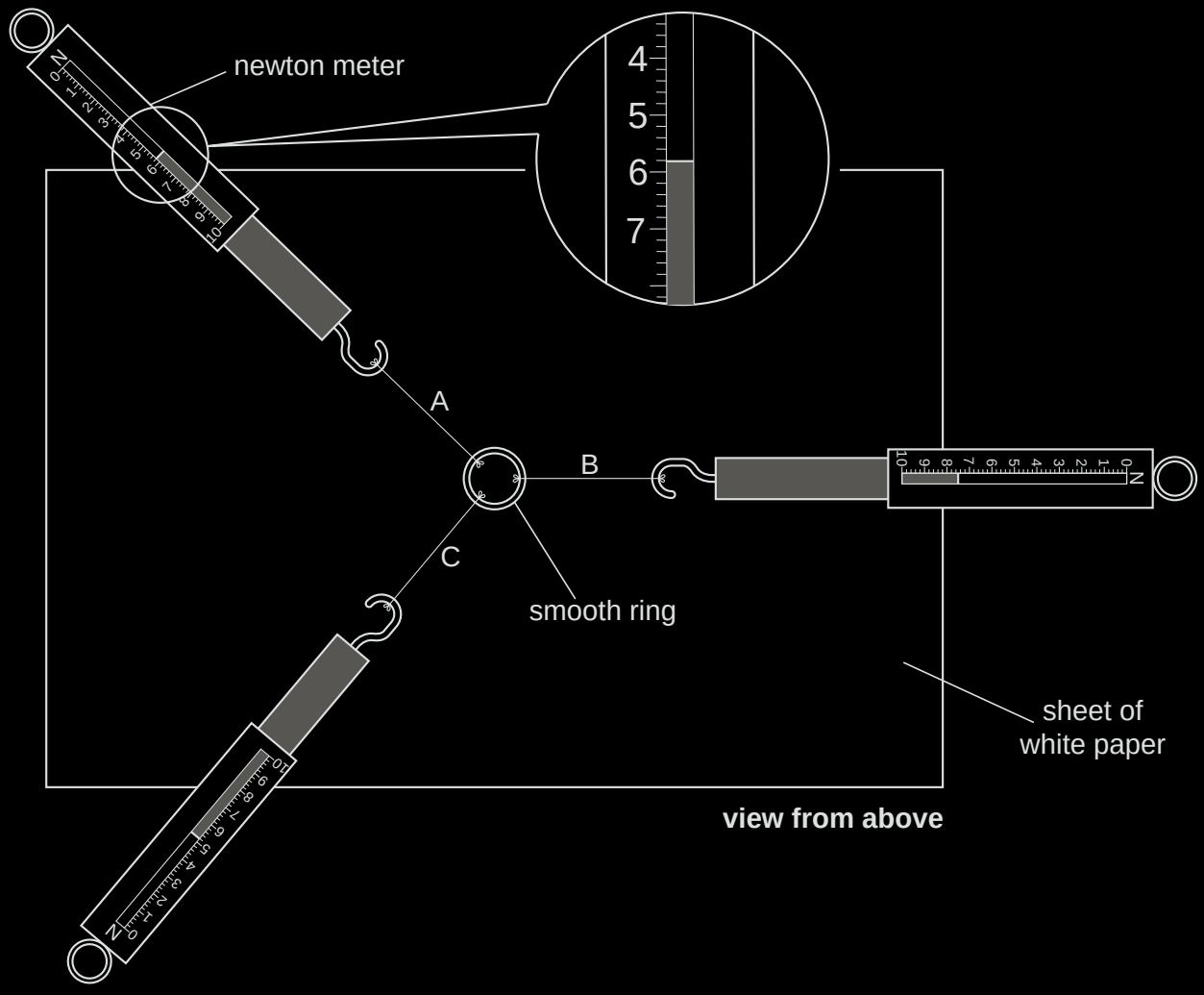



**Fig. 2.1**

The apparatus is used to find the vector sum of the forces in strings B and C.

The newton meters are pulled gently, then held in steady positions.

The positions of the strings A, B and C are marked with a pencil on the white paper.

**(a)** Explain why parallax error makes it difficult to mark the positions of the strings accurately.

.....

..... [1]

**(b)** Measure the angle between the strings A and B.

angle = ..... [1]

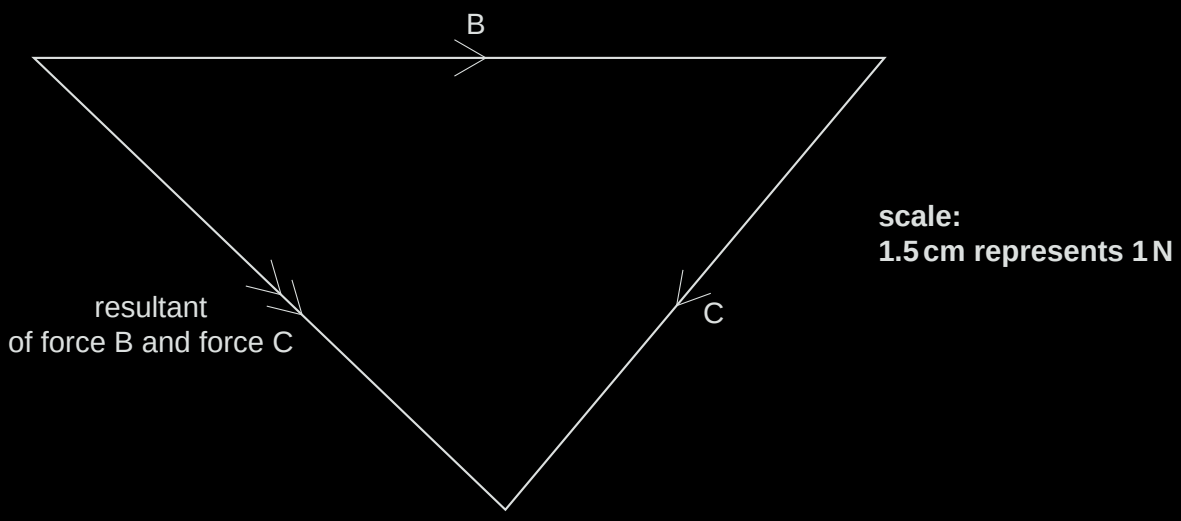



**Fig. 2.2**

Using the scale given, determine the magnitude of the resultant of force B and force C.

Show your working.

resultant force = ..... [1]

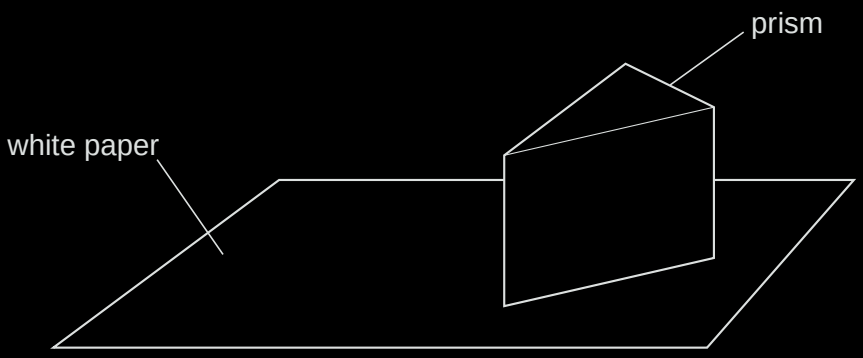



**Fig. 3.1**

Fig. 3.2 shows a ray of light from a ray box directed towards the prism.

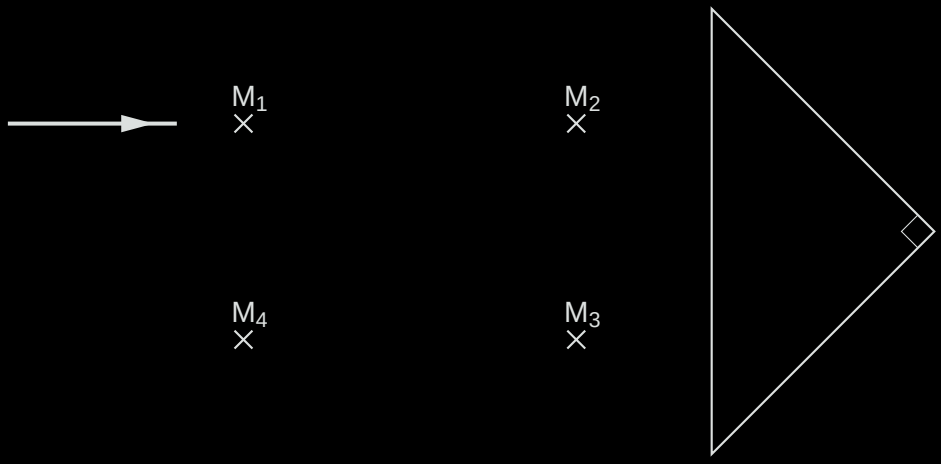



**Fig. 3.2**

A student makes two marks labelled $M_1$ and $M_2$ on the incident ray and two marks labelled $M_3$ and $M_4$ on the emergent ray.

**(a)** **(i)** On Fig. 3.2, continue the path of the ray through the prism. [1]

**(ii)** Describe the effect that the prism has on the ray of light.

.....

..... [1]

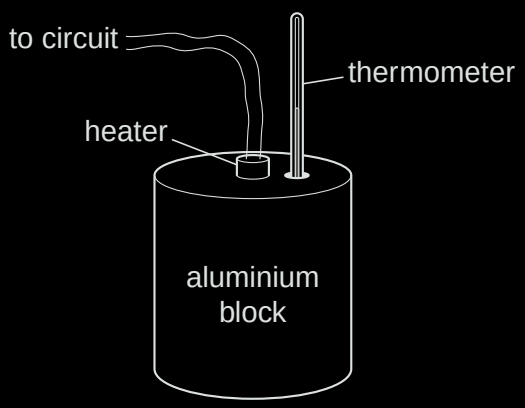



**Fig. 4.1**

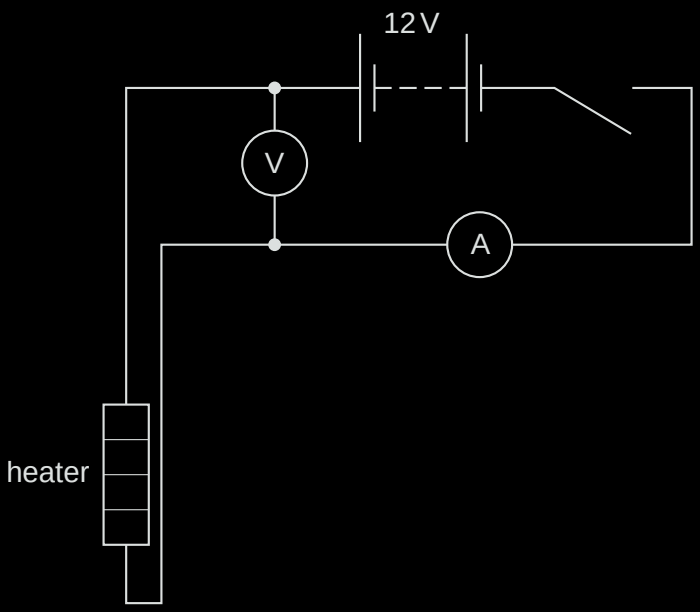



**Fig. 4.2**

A heater fits into a hole in the centre of the block and a thermometer fits into a second hole.

Fig. 4.2 is the circuit containing the heater.

When the switch is closed, the meters show steady readings of 11.6 V and 4.7 A.

**(a)** Fig. 4.3 shows the scales of the two meters.

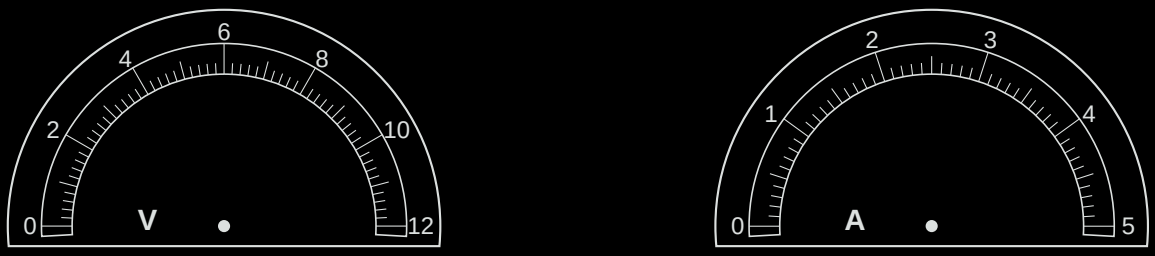



**Fig. 4.3**

On Fig. 4.3, draw pointers to show the readings on the meters after the switch has been closed.

[2]